THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 6: **Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A force of 240 N acts on an area of 0.4 square metre. The pressure is: 1

(a) 600 Pa **(b)** 240 Pa **(c)** 60 Pa **(d)** 96 Pa

2. Air pressure inside a bicycle tyre is greater than outside because: 1

(a) rubber is thick **(b)** the valve is open
(c) tyre is hot **(d)** more air is pumped in

3. A hurricane in the Atlantic and a typhoon in the Pacific are: 1

(a) earthquakes
(b) different types of storms
(c) the same type of storm with different names
(d) tornadoes

4. A brick of weight 20 N has faces of areas 0.1 square metre and 0.04 square metre. The maximum pressure it can exert when resting on a face is: 1

(a) 200 Pa **(b)** 500 Pa **(c)** 0.8 Pa **(d)** 2 Pa

5. A sealed bag of chips puffs up on a hilltop because: 1

(a) chips expand **(b)** outside pressure is lower than inside
(c) outside pressure is higher **(d)** air inside cools

6. A tornado forms mostly: 1

(a) in caves **(b)** over land under thunderclouds
(c) at the poles **(d)** over the deep sea always

7. A brick of weight 20 N has faces of areas 0.1 square metre and 0.04 square metre. The minimum pressure it can exert when resting on a face is: 1

(a) 2 Pa **(b)** 200 Pa **(c)** 0.2 Pa **(d)** 500 Pa

8. The sea breeze occurs because: 1

(a) land is cooler than sea in the daytime **(b)** sea is warmer at noon
(c) sea is frozen **(d)** land is warmer than sea in the daytime

9. The speed of a cyclone's wind is highest in the: 1

(a) eye **(b)** eye wall **(c)** the centre **(d)** outer edge only

10. A person of weight 600 N stands on both feet, each of area 0.01 square metre. The pressure on the ground is: 1

(a) 30000 Pa **(b)** 6000 Pa **(c)** 60000 Pa **(d)** 300 Pa

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A box exerts a force of 80 N on a floor over 0.2 square metre. Find the pressure. 2

12. How does a hot air balloon use the idea of air expansion? 2

13. Why do winds blow from high to low pressure? 2

14. Why does a person lying on a bed of nails not get hurt? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A girl of weight 400 N stands on one foot of area 0.02 square metre. Find the pressure and compare with standing on two feet. 3

16. Differentiate between a cyclone, a tornado and a thunderstorm (one point each). 3

17. How does a lift pump or dropper make use of air pressure? Explain one. 3

OR

The pressure at the bottom of a pond is more than at the top. Explain with an activity. 3

18. Describe how the Indian Meteorological Department's colour codes help people in a cyclone. 3

OR

Explain why tall buildings are fitted with lightning conductors in storm-prone areas. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A tractor with weight 36000 N has four wheels each in contact with 0.3 square metre of ground. Find the pressure on the ground and compare with a person of weight 600 N standing on feet of area 0.05 square metre in total. 5

OR

Describe the safety measures a community should take before, during and after a cyclone. 5

20. Explain pressure in liquids using the example of a dam and water jets. Mention the dependence on depth and direction. 5

OR

Explain why the pressure of air in a closed container increases when heated and how a pressure cooker works. 5

END OF QUESTION PAPER — Total: 40 Marks

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